

ORIGINAL RESEARCH

HEART FAILURE

Finerenone and Cardiovascular Outcomes According to Baseline Kidney Function in Patients With Heart Failure



The FINEARTS-HF Trial

Finnian R. Mc Causland, MBBCh, MSc,^a Safia Chatur, MD,^a Muthiah Vaduganathan, MD, MPH,^a Brian L. Claggett, PhD,^a Ian J. Kulac, MS,^a Akshay S. Desai, MD, MPH,^a Pardeep S. Jhund, MD, PhD,^b Alasdair D. Henderson, PhD,^b Katja Rohwedder, MD,^c Flaviana Amarante, MD,^c Meike Brinker, MD,^c Patrick Schloemer, PhD,^c Carolyn S.P. Lam, MD, PhD,^d John W. Ostrominski, MD,^a Michele Senni, MD,^e Sanjiv J. Shah, MD,^f Adriaan A. Voors, MD,^g Faiez Zannad, MD,^h Bertram Pitt, MD,ⁱ John J.V. McMurray, MD,^b Scott D. Solomon, MD^a

ABSTRACT

BACKGROUND Finerenone is known to reduce the risk of worsening heart failure (HF) events and cardiovascular (CV) death in patients with HF with mildly reduced or preserved ejection fraction.

OBJECTIVES The authors explored whether the benefit of finerenone among patients with HF with mildly reduced or preserved ejection fraction differs according to baseline measures of kidney function.

METHODS FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure) was a global, randomized clinical trial of finerenone vs placebo among patients with HF with mildly reduced or preserved ejection fraction. Finerenone was titrated to 20 mg/d if the estimated glomerular filtration rate (eGFR) was ≤ 60 mL/min/1.73 m² or 40 mg/d if eGFR was > 60 mL/min/1.73 m². The authors used a semiparametric proportional rates method, stratified by left ventricular ejection fraction ($< 60\%$; $\geq 60\%$) and region, to assess for differential treatment effects on the composite of total HF events and CV death according to the baseline eGFR (continuous and categories [≥ 60 mL/min/1.73 m², 45 to < 60 mL/min/1.73 m², < 45 mL/min/1.73 m²]) and urine albumin-creatinine ratio (UACR) (< 30 mg/g, 30 to < 300 mg/g, ≥ 300 mg/g).

RESULTS The effect of finerenone to reduce the primary endpoint of total HF events and CV death did not significantly differ according to baseline eGFR ($P_{\text{interaction}} = 0.14$ and 0.07 for continuous and categorical eGFR, respectively) with rate ratio 0.72 (95% CI: 0.59-0.88) for eGFR ≥ 60 mL/min/1.73 m², 0.83 (95% CI: 0.65-1.06) for eGFR 45 to < 60 mL/min/1.73 m², and 1.02 (95% CI: 0.83-1.26) for eGFR < 45 mL/min/1.73 m². The corresponding absolute event rates were 9.2 vs 12.5, 16.5 vs 19.9, and 28.0 vs 28.0 per 100 patient-years for finerenone vs placebo, respectively. Similar results were noted for total worsening HF events. Finerenone lowered the risk of the composite CV outcome similarly across baseline categories of UACR ($P_{\text{interaction}} = 0.48$).

CONCLUSIONS In the FINEARTS-HF trial (where the target dose of finerenone was determined by baseline kidney function), the effect of finerenone to reduce the composite of cardiovascular death and total HF events did not significantly differ across a range of baseline eGFR and UACR. (JACC Heart Fail. 2026;14:102778) © 2026 by the American College of Cardiology Foundation.

**ABBREVIATIONS
AND ACRONYMS****CKD** = chronic kidney disease**eGFR** = estimated glomerular filtration rate**HF** = heart failure**HFmrEF** = heart failure with mildly reduced ejection fraction**HFpEF** = heart failure with preserved ejection fraction**MRA** = mineralocorticoid receptor antagonist

Chronic kidney disease (CKD) is present in ~50% of patients with heart failure with mildly reduced ejection fraction (HFmrEF) or heart failure with preserved ejection fraction (HFpEF) and is associated with a higher risk of adverse cardiovascular (CV) events and death.¹ Furthermore, the presence of CKD may lead to hesitancy or failure to initiate or titrate some guideline-directed medical therapies for patients with heart failure (HF),² with concerns that include acute worsening of kidney function and/or hyperkalemia with treatments such as renin-angiotensin system inhibitors, mineralocorticoid receptor antagonists (MRAs), angiotensin receptor-neprilysin inhibitors (ARNIs), and sodium-glucose cotransporter 2 (SGLT2) inhibitors.³

Depending on the pharmacologic agent in question, prior trial-level analyses of patients with HFmrEF/HFpEF have observed a variety of differential treatment effects on reducing adverse CV events according to baseline kidney function. For example, in PARAGON-HF (Prospective Comparison of ARNI with ARB Global Outcomes in Heart Failure with Preserved Ejection Fraction), the CV benefits of sacubitril/valsartan (vs valsartan) appear to be more pronounced at lower levels of kidney function,⁴ whereas in DELIVER (Dapagliflozin Evaluation to Improve the LIVES of Patients With PReserved Ejection Fraction Heart Failure) and TOPCAT Americas (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist), the benefits of dapagliflozin and spironolactone (both vs placebo) appeared to be consistent across the spectrum of kidney function.^{5,6}

Finerenone, a nonsteroidal MRA, was recently reported to lower the risk of total HF events and CV death among patients with HFmrEF/HFpEF in the FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure) trial.⁷ Given the heterogeneity of differential treatment effects in prior trials of

patients with HFmrEF/HFpEF and the estimated glomerular filtration rate (eGFR)-based dosing strategy in FINEARTS-HF, in these prespecified analyses, we explore if the CV benefits of finerenone vs placebo differ according to the baseline kidney function.

METHODS

The trial design, analysis plan, and primary results of the FINEARTS-HF trial have been published.^{7,8} The trial protocol was approved by the ethics committees or institutional review boards at all participating centers and all patients have provided written informed consent. Trial oversight was provided by an independent data and safety monitoring committee. The first author wrote the first draft and all authors contributed to revisions and made the collective decision to submit the manuscript for publication.

STUDY PATIENTS. Briefly, FINEARTS-HF enrolled patients ≥ 40 years of age with symptomatic HF, left ventricular ejection fraction (LVEF) $\geq 40\%$, evidence of structural heart disease (left ventricular hypertrophy or left atrial enlargement within 12 months), and elevated serum concentration of natriuretic peptides. Notable exclusion criteria included: eGFR < 25 mL/min/1.73 m²; serum potassium > 5 mmol/L; symptomatic hypotension (systolic blood pressure < 90 mm Hg); treatment with more than one angiotensin-converting enzyme inhibitor, angiotensin receptor blocker; or ARNI at randomization. Patients were randomized in a 1:1 fashion to finerenone or matching placebo. The initial dose was 10 mg daily for those with baseline eGFR ≤ 60 mL/min/1.73 m² (titrated to a maximum of 20 mg daily) or 20 mg daily for those with baseline eGFR > 60 mL/min/1.73 m² (titrated to maximum of 40 mg daily).

PRIMARY CV OUTCOME. The prespecified primary outcome was the composite of total (first and recurrent) HF events and CV death. Additional outcomes considered included total and first worsening HF events, CV death, and all-cause death.

From the ^aBrigham and Women's Hospital, Harvard Medical School, Boston, Massachusetts, United States; ^bUniversity of Glasgow, Glasgow, Scotland, United Kingdom; ^cBayer, Berlin, Germany; ^dNational Heart Centre Singapore and Duke-National University of Singapore, Singapore; ^eUniversity of Milano-Bicocca ASST Papa Giovanni XXIII Hospital, Bergamo, Italy; ^fNorthwestern University Feinberg School of Medicine, Chicago, Illinois, United States; ^gUniversity of Groningen, Groningen, the Netherlands; ^hUniversity of Lorraine, Nancy, France; and the ⁱUniversity of Michigan, Ann Arbor, Michigan, United States. Biykem Bozkurt, MD, PhD, served as acting Editor-in-Chief and main adjudicator for this paper.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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SAFETY OUTCOMES. For the present analysis, any discontinuation of the randomized treatment, discontinuation caused by an adverse event (AE), and discontinuation caused by a serious adverse event (SAE) were examined. These were analyzed as events that occurred among patients who had received at least 1 dose of finerenone or placebo up until the end of study data collection. Discontinuation caused by treatment-emergent investigator-reported hyperkalemia and acute kidney injury were too infrequent to allow meaningful analyses ($n = 18$ and $n = 3$, respectively).

RISK-BENEFIT ANALYSIS. The trade-offs in absolute risks and benefits of finerenone across the spectrum of eGFR were evaluated by estimating the annual incidence rate differences (IRDs) of the primary efficacy outcome (time to first worsening HF event or CV death), as well as safety outcomes: 1) including AEs leading to drug discontinuation; and 2) any hyperkalemia >5.5 mmol/L—not necessarily leading to discontinuation according to baseline eGFR categories. The relationship between treatment, efficacy, safety estimates, and baseline kidney function as a continuous variable was assessed using restricted cubic splines.

BASELINE KIDNEY FUNCTION. Baseline kidney function was assessed by eGFR, using the CKD-EPI formula, as outlined in the original study protocol. For these analyses, baseline eGFR was considered as both a continuous and categorical variable (≥ 60 mL/min/1.73 m², 45 to <60 mL/min/1.73 m², and <45 mL/min/1.73 m²).

STATISTICAL ANALYSIS. Data were reported as mean $\pm$ SD when normally distributed, median (Q1-Q3) when non-normally distributed, and as frequencies and percentages for categorical data. Baseline comparisons were made according to categories of eGFR using tests for trend based on linear regression, the chi square trend test, and the Cuzick nonparametric trend test, as appropriate. Exposure-adjusted incidence rates and 95% CIs were calculated per 100 patient-years of follow-up (calculated as the number of patients with an event divided by the total exposure time of the population). Event rates across the spectrum of eGFR for time to the composite of first CV death or HF hospitalization, CV death, first HF hospitalization, all-cause mortality, and safety events were evaluated using Poisson regression with eGFR and urine albumin-creatinine ratio (UACR) expressed using restricted cubic splines with 3 knots. Incidence rates for recurrent event endpoints additionally included robust variance estimates. Knots were assigned automatically

according to data distribution.⁹ Nonlinearity was assessed by comparison of the spline model with a linear model using likelihood ratio testing.

Semiparametric methods, stratified by geographic region and baseline LVEF ($<60\%$ or $\geq 60\%$), were used to estimate the treatment effect on the primary composite outcome of total HF events or CV death and the outcome of total HF events alone. Cox proportional hazards models, stratified by geographic region and LVEF ($<60\%$ or $\geq 60\%$), were fit to assess the effect of finerenone vs placebo on time to first event outcomes (CV death and all-cause death). Interaction terms were added to models to explore for differential treatment effects according to the baseline eGFR (considered as both a continuous [linear] and categorical variable). Nonlinearity in treatment effects across the range of eGFR was also assessed using restricted cubic spline models.

In sensitivity analyses, as mortality may preclude the development of HF or other cause-specific death, a competing risk model was fit according to the method of Fine and Gray¹⁰ to estimate the cumulative incidence function for time-to-first-event outcomes. In these models, all-cause mortality was considered a competing risk for HF outcomes, whereas non-CV death was considered as the competing risk for CV death. Some recent trials of patients with HFmrEF/HFpEF have noted issues with potential violation of the proportional hazards assumption.¹¹ Therefore, to address the potential influence of nonproportional hazards, a model-free approach based on a comparison of areas under the Kaplan-Meier curves up to 3 years of follow-up was performed. Consistent with the primary trial, all efficacy analyses for the present paper were performed according to the intention-to-treat principle. To address the concerns that adverse clinical outcomes could occur more frequently after discontinuation of finerenone vs placebo, on-treatment analyses were also performed considering the period from the date of initiation of randomized treatment through 3 days after the end date of randomized treatment.

Analogous approaches were used to examine the association of baseline albuminuria (<30 mg/g, 30 to <300 mg/g, ≥ 300 mg/g) with outcomes of interest. All present analyses were performed at an alpha level of 0.05, without correction for multiple hypothesis testing, using Stata MP (version 18.0, Stata Corp).

RESULTS

BASELINE CHARACTERISTICS. The mean baseline eGFR was 62 ± 20 mL/min/1.73 m². Overall, 3,113 (52%) had an eGFR ≥ 60 mL/min/1.73 m², 1,556 (26%)

TABLE 1 Baseline Characteristics According to Category of CKD and Randomized Treatment Assignment^a

	eGFR ≥60 mL/min/1.73 m ²		eGFR 45 to <60 mL/min/1.73 m ²		eGFR <45 mL/min/1.73 m ²	
	Placebo (n = 1,561)	Finerenone (n = 1,552)	Placebo (n = 754)	Finerenone (n = 802)	Placebo (n = 683)	Finerenone (n = 649)
Age, y	68 ± 10	69 ± 10	75 ± 8	74 ± 8	77 ± 8	77 ± 8
Female	643 (41.2)	637 (41.0)	375 (49.7)	400 (49.9)	359 (52.6)	318 (49.0)
Race ^b						
Asian	255 (16.3)	252 (16.2)	118 (15.6)	122 (15.2)	126 (18.4)	123 (19.0)
Black	22 (1.4)	28 (1.8)	9 (1.2)	15 (1.9)	8 (1.2)	6 (0.9)
Other	39 (2.5)	47 (3.0)	27 (3.6)	22 (2.7)	25 (3.7)	22 (3.4)
White	1,245 (79.8)	1,225 (78.9)	600 (79.6)	643 (80.2)	524 (76.7)	498 (76.7)
Geographic region						
Asia	249 (16.0)	250 (16.1)	117 (15.5)	121 (15.1)	124 (18.2)	122 (18.8)
Eastern Europe	794 (50.9)	762 (49.1)	324 (43.0)	346 (43.1)	203 (29.7)	221 (34.1)
Latin America	175 (11.2)	179 (11.5)	67 (8.9)	77 (9.6)	77 (11.3)	66 (10.2)
North America	94 (6.0)	96 (6.2)	76 (10.1)	80 (10.0)	66 (9.7)	59 (9.1)
Western Europe, Oceania, and others	249 (16.0)	265 (17.1)	170 (22.5)	178 (22.2)	213 (31.2)	181 (27.9)
Any prior hospitalization for HF	913 (58.5)	905 (58.3)	453 (60.1)	474 (59.1)	456 (66.8)	418 (64.4)
Systolic blood pressure, mm Hg	130 ± 15	130 ± 15	130 ± 16	129 ± 15	128 ± 16	130 ± 16
BMI, ^c kg/m ²	30.2 ± 6.1	29.7 ± 6.0	30.2 ± 6.3	30.4 ± 6.3	29.4 ± 6.0	29.7 ± 6.1
Serum creatinine, mg/dL	0.9 ± 0.2	0.9 ± 0.2	1.2 ± 0.2	1.2 ± 0.2	1.6 ± 0.5	1.6 ± 0.3
eGFR, mL/min/1.73 m ²	78 ± 12	77 ± 12	53 ± 4	53 ± 4	36 ± 6	36 ± 6
UACR, mg/g	14 (6-45)	14 (5-45)	20 (7-77)	19 (8-75)	31 (11-148)	36 (11-176)
Serum potassium, mmol/L	4.3 ± 0.5	4.4 ± 0.4	4.4 ± 0.5	4.4 ± 0.5	4.4 ± 0.5	4.4 ± 0.5
LVEF, %	52 ± 8	52 ± 8	53 ± 8	53 ± 8	53 ± 8	54 ± 8
NT-proBNP, pg/mL	766 (338-1,486)	844 (372-1,529)	1,219 (535-2,339)	1,152 (536-2,030)	1,581 (790-3,025)	1,638 (784-2,946)
History of hypertension	1,370 (87.8)	1,319 (85.0)	687 (91.1)	731 (91.1)	628 (91.9)	590 (90.9)
History of diabetes	583 (37.3)	556 (35.8)	301 (39.9)	338 (42.1)	338 (49.5)	323 (49.8)
History of stroke	161 (10.3)	154 (9.9)	88 (11.7)	105 (13.1)	104 (15.2)	96 (14.8)
History of myocardial infarction	411 (26.3)	446 (28.7)	187 (24.8)	188 (23.4)	159 (23.3)	150 (23.1)
Beta-blocker	1,342 (86.0)	1,316 (84.8)	643 (85.3)	688 (85.8)	569 (83.3)	537 (82.7)
ACEI	616 (39.5)	613 (39.5)	257 (34.1)	275 (34.3)	199 (29.1)	195 (30.0)
ARB	521 (33.4)	503 (32.4)	287 (38.1)	316 (39.4)	247 (36.2)	228 (35.1)
ARNI	154 (9.9)	141 (9.1)	56 (7.4)	60 (7.5)	47 (6.9)	55 (8.5)
SGLT2 inhibitor	188 (12.0)	180 (11.6)	103 (13.7)	116 (14.5)	133 (19.5)	97 (14.9)
Loop diuretic	1,329 (85.1)	1,329 (85.6)	662 (87.8)	685 (85.4)	630 (92.2)	604 (93.1)
Thiazide diuretic	225 (14.4)	231 (14.9)	110 (14.6)	130 (16.2)	67 (9.8)	68 (10.5)

Values are mean ± SD, n (%), and median (Q1-Q3). ^aPercentages may not be of total 100% because of rounding. ^bRace was self-reported by the patient. ^cSignificant differences were noted for BMI and baseline eGFR in eGFR category >60 mL/min/1.73 m²; systolic blood pressure and baseline SGLT2 inhibitor use in eGFR category <45 mL/min/1.73 m².

ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin-receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; BMI = body mass index; CKD = chronic kidney disease; eGFR = estimated glomerular filtration rate; HF = heart failure; LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-B-type natriuretic peptide; SGLT2 = sodium-glucose cotransporter 2; UACR = urine albumin-creatinine ratio.

had an eGFR 45 to <60 mL/min/1.73 m², and 1,332 (22%) had an eGFR <45 mL/min/1.73 m². Patients with lower baseline eGFR tended to be older, female, of Asian descent, with lower body mass index and systolic blood pressure, but higher serum potassium and UACR. Those with lower eGFR were more likely to have had a prior HF hospitalization, higher LVEF and N-terminal pro-B-type natriuretic peptide, and higher frequency of hypertension, diabetes mellitus, and stroke. They were less likely to have been prescribed an angiotensin-converting enzyme inhibitor, ARNI, or thiazide diuretic, but more likely to have

been prescribed an angiotensin receptor blocker, SGLT2 inhibitor, or loop diuretic (**Table 1**).

During the trial, the mean prescribed dose of finerenone/placebo was 32 ± 9 mg for those with baseline eGFR ≥60 mL/min/1.73 m², 20 ± 8 mg for those with eGFR 45 to <60 mL/min/1.73 m², and 15 ± 5 mg for those with eGFR <45 mL/min/1.73 m².

TREATMENT EFFECT ON THE CV COMPOSITE (TOTAL HF EVENTS OR CV DEATH) ACCORDING TO BASELINE eGFR. Overall, the rate of the CV composite outcome was higher at lower levels of eGFR

TABLE 2 Treatment Effect on CV Outcomes According to Baseline eGFR Categories

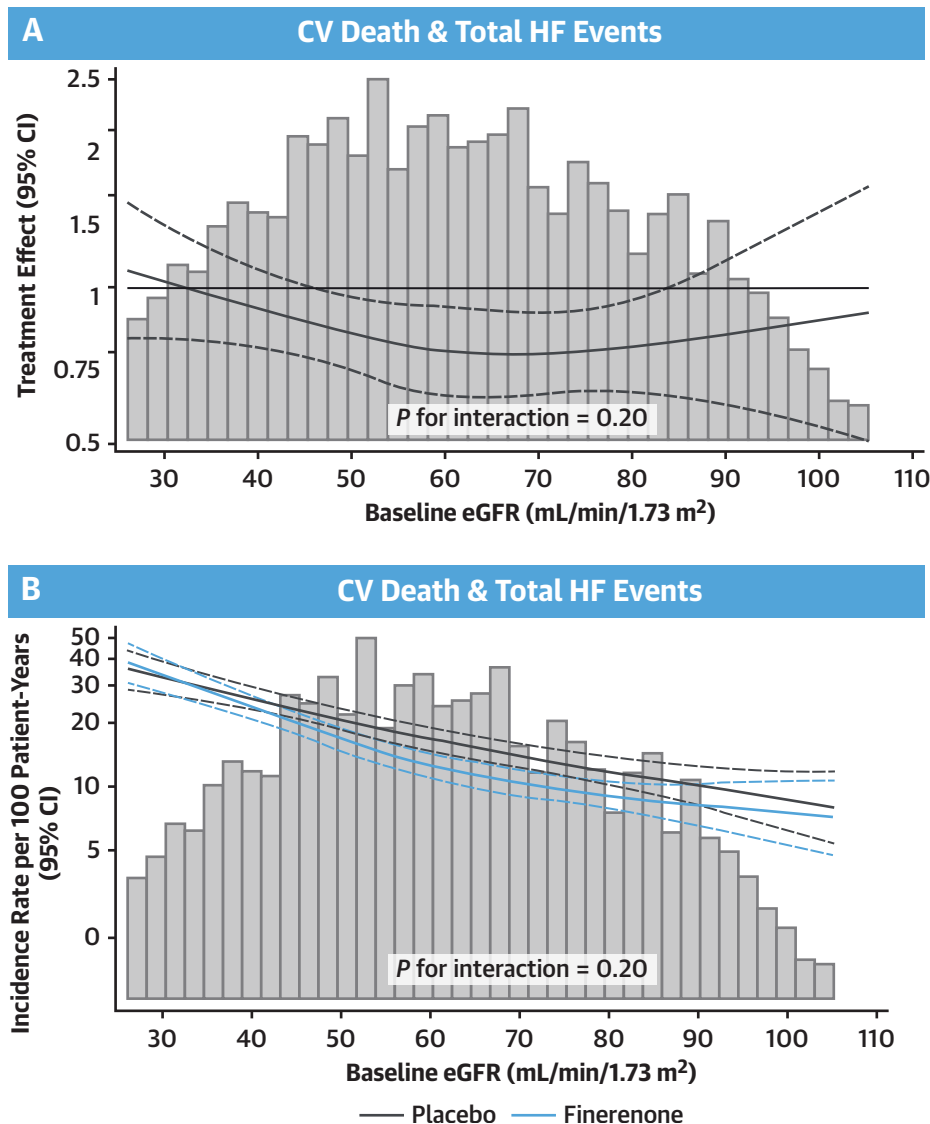
	eGFR ≥60 mL/min/1.73 m ²		eGFR 45 to <60 mL/min/1.73 m ²		eGFR <45 mL/min/1.73 m ²		P _{Interaction}
	Placebo	Finerenone	Placebo	Finerenone	Placebo	Finerenone	
Total HF events and CV death							
Events/patients	487/1,561	356/1,552	359/754	320/802	437/683	407/649	
Rate/100 py (95% CI)	12.5 (10.8-14.2)	9.2 (7.8-10.5)	19.9 (16.6-23.3)	16.5 (13.4- 19.7)	28.0 (23.9-32.2)	28.0 (23.6-32.4)	
RR (95% CI)	0.72 (0.59-0.88)		0.83 (0.65-1.06)		1.02 (0.83-1.26)		0.07
Absolute rate difference	−3.4 (−5.6 to −1.2)		−3.4 (−8.0 to 1.2)		−0.1 (−6.1 to 6.0)		0.60
Total HF events							
Events/patients	379/1561	266/1,552	291/754	250/802	354/683	326/649	
Rate/100 py (95% CI)	9.7 (8.2-11.3)	6.8 (5.6-8.1)	16.2 (13.0- 19.3)	12.9 (10.1-15.7)	22.7 (19.0-26.5)	22.4 (18.5-26.3)	
RR (95% CI)	0.69 (0.54-0.88)		0.80 (0.60-1.06)		1.01 (0.80-1.28)		0.08
Absolute rate difference	−2.9 (−4.9 to −0.9)		−3.2 (−7.5 to 1.0)		−0.3 (−5.7 to 5.1)		0.65
First worsening HF event							
Events/patients (%)	223/1,561 (14.3)	166/1,552 (10.7)	153/754 (20.3)	138/802 (17.2)	197/683 (28.8)	175/649 (27.0)	
Rate/100 py (95% CI)	6.2 (5.4-7.0)	4.5 (3.8-5.2)	9.5 (8.0-11.1)	7.8 (6.5-9.1)	14.8 (12.7-17.0)	14.0 (11.9-16.2)	
HR (95% CI)	0.72 (0.59-0.88)		0.81 (0.64-1.02)		0.98 (0.80-1.20)		0.12
Absolute rate difference	−1.7 (−2.8 to −0.6)		−1.7 (−3.8 to 0.3)		−0.8 (−3.9 to 2.2)		0.87
CV death							
Events/patients (%)	108/1,561 (6.9)	91/1,552 (5.9)	69/754 (9.2)	70/802 (8.7)	83/683 (12.2)	81/649 (12.5)	
Rate/100 py (95% CI)	2.8 (2.3-3.3)	2.3 (1.9-2.8)	3.8 (2.9-4.7)	3.6 (2.8-4.5)	5.3 (4.2-6.5)	5.6 (4.4-6.8)	
HR (95% CI)	0.84 (0.64-1.12)		0.96 (0.69-1.34)		1.06 (0.78-1.44)		0.58
Absolute rate difference	−0.4 (−1.1 to 0.3)		−0.2 (−1.5 to 1.0)		0.2 (−1.4 to 1.9)		0.76
All-cause death							
Events/patients (%)	197/1,561 (12.6)	172/1,552 (11.1)	152/754 (20.2)	134/802 (16.7)	173/683 (25.3)	185/649 (28.5)	
Rate/100 py (95% CI)	5.0 (4.3-5.7)	4.4 (3.7-5.1)	8.4 (7.1-9.7)	6.9 (5.7-8.1)	11.1 (9.4-12.7)	12.6 (10.8-14.4)	
HR (95% CI)	0.88 (0.71-1.08)		0.82 (0.65-1.04)		1.15 (0.93-1.41)		0.07
Absolute rate difference	−0.6 (−1.6 to 0.3)		−1.5 (−3.3 to 0.2)		1.6 (−0.9 to 4.0)		0.13

CV = cardiovascular; py = person-years; RR = rate ratio; other abbreviation as in Table 1.

(Table 2, Central Illustration). Among those with baseline eGFR ≥ 60 mL/min/1.73 m², there were 487 total events among 1,561 patients randomized to placebo and 356 total events among 1,552 patients randomized to finerenone (rate ratio [RR]: 0.72; 95% CI: 0.59-0.88). Among those with baseline eGFR 45 to <60 mL/min/1.73 m², there were 359 total events among 754 patients randomized to placebo and 320 total events among 802 patients randomized to finerenone (RR: 0.83; 95% CI: 0.65-1.06). Among those with baseline eGFR <45 mL/min/1.73 m², there were 437 total events among 683 patients randomized to placebo and 407 total events among 649 patients randomized to finerenone (RR: 1.02; 95% CI: 0.83-1.26). Although the effect estimates were attenuated with lower eGFR categories, there was no clear evidence of statistical heterogeneity (*P*_{interaction} = 0.07). Similar findings were observed when eGFR was considered to be a continuous variable (Central Illustration). Absolute rate differences were similar across categories of baseline eGFR (*P*_{interaction} = 0.60) (Table 2).

TREATMENT EFFECT ON TOTAL HF EVENTS AND FIRST OCCURRENCE OF WORSENING HF EVENTS ACCORDING TO BASELINE eGFR. The frequency of total HF events was higher across lower eGFR categories. Although the magnitude of risk reduction was numerically greatest among those with higher baseline eGFR (RR: 0.69; 95% CI: 0.54-0.88 for those with ≥ 60 mL/min/1.73 m²; RR: 0.80; 95% CI: 0.60-1.06 for those with eGFR 45 to <60 mL/min/1.73 m²; RR: 1.01; 95% CI: 0.80-1.28 for those with eGFR <45 mL/min/1.73 m²), there was no clear evidence of statistical heterogeneity (*P*_{interaction} = 0.08) (Table 2). Absolute rate differences were similar across categories of baseline eGFR (*P*_{interaction} = 0.65) (Table 2). Similar magnitudes and patterns of treatment effects were observed when the first HF event was considered as the outcome of interest (Table 2).

TREATMENT EFFECT ON CV DEATH AND ALL-CAUSE DEATH ACCORDING TO BASELINE eGFR. Rates of CV death and all-cause death were higher among

CENTRAL ILLUSTRATION Treatment Effect and Incidence Rates for Cardiovascular Death and Total Heart Failure Events Across Baseline eGFR

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(A) The treatment effect for finerenone vs placebo on the composite of cardiovascular (CV) death and total heart failure (HF) events according to baseline estimated glomerular filtration rate (eGFR). The solid line represents the treatment effect (rate ratio) for finerenone vs placebo; the dashed lines represent the 95% CIs; the histogram reflects the distribution of patients across the baseline eGFR. (B) The incidence rates for composite of CV death and total HF events according to baseline eGFR and randomized treatment assignment. The solid lines represent the incidence rate per 100 patient-years for finerenone (blue) and placebo (black); the dashed lines represent the 95% CIs; the histogram reflects the distribution of patients across the baseline eGFR.

those in lower eGFR categories. No statistically significant reductions in CV or all-cause mortality were noted across eGFR categories for finerenone vs placebo ($P_{\text{interaction}} = 0.58$ and 0.07 , respectively) (Table 2).

SENSITIVITY ANALYSES: COMPETING RISKS, MODEL-FREE APPROACHES, AND ON-TREATMENT ANALYSES. For the outcomes of the first occurrence of HF event or CV death, the patterns of association with competing risk analyses and model-free analyses

TABLE 3 Sensitivity Analyses Using Competing Risk and Nonproportional Hazards Models for Time to First Event

	eGFR ≥60 mL/min/1.73 m ²		eGFR 45 to <60 mL/min/1.73 m ²		eGFR <45 mL/min/1.73 m ²		P _{interaction}
	Placebo	Finerenone	Placebo	Finerenone	Placebo	Finerenone	
First worsening HF event							
Events/patients (%)	223/1,561 (14.3)	166/1,552 (10.7)	153/754 (20.3)	138/802 (17.2)	197/683 (28.8)	175/649 (27.0)	0.19
Rate/100 py (95% CI)	6.2 (5.4-7.1)	4.5 (3.9-5.3)	9.5 (8.1-11.1)	7.8 (6.6-9.2)	14.8 (12.9-17.1)	14.0 (12.1-16.2)	
Competing risk, sHR (95% CI)	0.72 (0.59-0.88)		0.83 (0.66-1.05)		0.94 (0.76-1.15)		
Model-free ratio (95% CI)	0.67 (0.53-0.84)		0.82 (0.63-1.07)		0.88 (0.72-1.09)		
CV death							
Events/patients (%)	108/1,561 (6.9)	91/1,552 (5.9)	69/754 (9.2)	70/802 (8.7)	83/683 (12.2)	81/649 (12.5)	0.64
Rate/100 py (95% CI)	2.8 (2.3-3.4)	2.3 (1.9-2.9)	3.8 (3.0-4.9)	3.6 (2.9-4.6)	5.3 (4.3-6.6)	5.6 (4.5-6.9)	
Competing risk, sHR (95% CI)	0.84 (0.64-1.11)		0.97 (0.69-1.35)		1.03 (0.76-1.39)		
Model-free ratio (95% CI)	0.83 (0.58-1.19)		0.96 (0.63-1.48)		1.11 (0.75-1.65)		
Competing risk models use the Fine and Gray approach, where all-cause mortality was considered a competing risk for HF outcomes, whereas non-CV death was the competing risk for CV death. The model-free approach addresses the potential influence of nonproportional hazards and is based on a comparison of areas under the Kaplan-Meier curves, up to 3 years of follow-up.							
sHR = subdistribution HR; other abbreviations as in Table 2.							

that addressed potential nonproportional hazards were similar to those observed in the main analyses (Table 3). For all outcomes considered, the patterns of treatment effect for on-treatment analyses were similar to those from the intention-to-treat analyses, with numerically larger effects observed for those with higher baseline eGFR, but without clear evidence of statistical heterogeneity (Supplemental Table 2). Absolute rate differences were similar across baseline categories of eGFR (Supplemental Table 2).

CLINICAL OUTCOMES ACCORDING TO BASELINE UACR. For all outcomes considered, the frequency of events was higher among those with higher baseline UACR. The benefits of finerenone vs placebo were consistent across UACR categories for the composite cardiovascular outcome, total HF events, and first HF event ($P_{\text{interaction}} > 0.10$ for all) (Table 4).

No statistically significant reductions in CV or all-cause mortality were noted across UACR categories for finerenone vs placebo, with no evidence of heterogeneity ($P_{\text{interaction}} = 0.17$ and 0.82 , respectively) (Table 4).

SAFETY OUTCOMES ACCORDING TO CATEGORIES OF BASELINE eGFR AND UACR. The overall rate of randomized treatment discontinuation was 21% and occurred more frequently among those with lower baseline eGFR and higher UACR. The frequency of any discontinuation was numerically higher for finerenone vs placebo among those in the lowest baseline eGFR category. However, AEs and SAEs requiring discontinuation were similar between those randomized to finerenone and placebo across baseline categories of eGFR and UACR ($P_{\text{interaction}} > 0.20$ for all) (Table 5).

BALANCE OF RISK AND BENEFIT OF FINERENONE ACROSS THE SPECTRUM OF eGFR AND UACR.

The annual IRD for finerenone relative to placebo on the primary efficacy outcome was consistent across all eGFR categories (IRD per 100 patient-years, where positive values represent events prevented: $+0.4$ [95% CI: -2.8 to 3.6] for eGFR <45 mL/min/1.73 m²; $+1.9$ [95% CI: -0.4 to 4.2] for eGFR 45 to <60 mL/min/1.73 m²; and $+1.9$ [95% CI: 0.7 - 3.1] for eGFR ≥ 60 mL/min/1.73 m²; $P_{\text{interaction}} = 0.67$) (Figure 1). The annual IRD for finerenone relative to placebo for AEs leading to drug discontinuation was also consistent across eGFR categories (IRD per 100 patient-years, where positive values represent excess events: $+1.4$ [95% CI: -0.2 to 2.9] for eGFR <45 mL/min/1.73 m²; -0.2 [95% CI: -1.1 to 0.8] for eGFR 45 to >60 mL/min/1.73 m²; $+0.2$ [95% CI: -0.3 to 0.7] for eGFR ≥ 60 mL/min/1.73 m²; $P_{\text{interaction}} = 0.28$) (Figure 1). Absolute risks of any hyperkalemia (>5.5 mmol/L) appeared to be amplified at lower eGFR: (IRD per 100 patient-years, where positive values represent excess events: $+6.0$ [95% CI: 3.8 - 8.0] for eGFR <45 mL/min/1.73 m²; $+3.7$ [95% CI: 2.2 - 5.3] for eGFR 45 to <60 mL/min/1.73 m²; $+2.7$ [95% CI: 1.8 - 3.6] for eGFR ≥ 60 mL/min/1.73 m²; $P_{\text{interaction}} = 0.02$).

The annual IRD for finerenone relative to placebo was similar across categories for the primary efficacy outcome and for AEs leading to drug discontinuation, but absolute risks for any hyperkalemia were higher for higher baseline UACR (Supplemental Figure 2).

DISCUSSION

Among patients with HFmrEF/HFpEF enrolled in the FINEARTS-HF trial, the effect of finerenone vs

TABLE 4 Treatment Effect on CV Outcomes According to Baseline UACR Categories

	UACR <30 mg/g		UACR 30 to <300 mg/g		UACR ≥300 mg/g		<i>P</i> _{Interaction}
	Placebo	Finerenone	Placebo	Finerenone	Placebo	Finerenone	
Total HF events and CV death							
Events/patients	518/1,746	429/1,765	477/868	427/844	228/282	174/292	
Rate/100 py (95% CI)	12.0 (10.5-13.5)	9.7 (8.3-11.1)	23.0 (19.7-26.4)	21.3 (17.8-24.8)	36.2 (28.3-44.1)	27.2 (21.3-33.2)	
RR (95% CI)	0.81 (0.67-0.97)		0.93 (0.75-1.16)		0.76 (0.57-1.03)		0.48
Absolute rate difference	−2.3 (−4.4 to −0.2)		−1.7 (−6.6 to 3.1)		−9.0 (−18.9 to 0.9)		0.40
Total HF events							
Events/patients	408/1,746	322/1,765	389/868	337/844	178/282	142/292	
Rate/100 py (95% CI)	9.5 (8.1-10.8)	7.3 (6.1-8.5)	18.8 (15.7-21.9)	16.8 (13.7-20.0)	28.3 (21.0-35.5)	22.2 (16.8-27.6)	
RR (95% CI)	0.76 (0.61-0.95)		0.90 (0.70-1.14)		0.79 (0.56-1.11)		0.63
Absolute rate difference	−2.2 (−4.0 to −0.3)		−2.0 (−6.3 to 2.4)		−6.1 (−15.1 to 3.0)		0.70
First worsening HF event							
Events/patients (%)	252/1,746 (14.4)	196/1,765 (11.1)	207/868 (23.8)	176/844 (20.9)	89/282 (31.6)	82/292 (28.1)	
Rate/100 py (95% CI)	6.3 (5.5-7.1)	4.7 (4.0-5.4)	11.5 (9.8-13.1)	9.9 (8.4-11.4)	17.4 (13.5-21.2)	15.3 (11.8-18.8)	
HR (95% CI)	0.74 (0.61-0.89)		0.87 (0.71-1.06)		0.87 (0.64-1.17)		0.46
Absolute rate difference	−1.6 (−2.6 to −0.6)		−1.6 (−3.8 to 0.6)		−2.1 (−7.3 to 3.1)		0.99
CV death							
Events/patients (%)	111/1,746 (6.4)	108/1,765 (6.1)	88/868 (10.1)	90/844 (10.7)	50/282 (17.7)	32/292 (11.0)	
Rate/100 py (95% CI)	2.6 (2.1-3.1)	2.5 (2.0-2.9)	4.2 (3.4-5.1)	4.5 (3.6-5.4)	7.9 (5.7-10.1)	5.0 (3.3-6.7)	
HR (95% CI)	0.96 (0.73-1.25)		1.09 (0.81-1.46)		0.67 (0.43-1.05)		0.17
Absolute rate difference	−0.1 (−0.8 to 0.5)		0.2 (−1.0 to 1.5)		−2.9 (−5.7 to −0.1)		0.12
All-cause death							
Events/patients (%)	243/1,746 (13.9)	223/1,765 (12.6)	174/868 (20.0)	164/844 (19.4)	85/282 (30.1)	81/292 (27.7)	
Rate/100 py (95% CI)	5.6 (4.9-6.3)	5.0 (4.4-5.7)	8.3 (7.1-9.6)	8.1 (6.9-9.4)	13.5 (10.6-16.3)	12.6 (9.8-15.4)	
HR (95% CI)	0.90 (0.75-1.07)		0.99 (0.80-1.23)		0.96 (0.71-1.32)		0.82
Absolute rate difference	−0.6 (−1.6 to 0.4)		−0.2 (−1.9 to 1.5)		−0.9 (−4.9 to 3.1)		0.92

Abbreviations as in Tables 1 and 2.

placebo to reduce the composite of CV death and total HF events did not significantly differ across separate metrics of baseline kidney function (both eGFR and UACR). Although the absolute risks of hyperkalemia were amplified at lower eGFR, the risks of AEs leading to discontinuation were similar across the spectrum of kidney function.

CKD, defined as eGFR <60 mL/min/1.73 m², is a common comorbid condition among patients with HFmrEF/HFpEF in recent outcomes trials and is a potent risk factor for adverse CV outcomes, including HF, compared to the absence of CKD.^{4-6,12-15} However, lower levels of kidney function are often accompanied by concerns related to a higher frequency of adverse outcomes with some pharmacologic interventions, potentially contributing to the under-treatment of these high-risk patients.² In DELIVER (dapagliflozin vs placebo on CV death or first HF event) and EMPEROR-Preserved (empagliflozin vs placebo on CV death and first HF hospitalization), the beneficial effects of SGLT2 inhibitor to lower the primary composite outcome were consistent across baseline eGFR,

without major concerns for excess safety risks.^{5,12,13} Among patients in the PARAGON-HF trial (sacubitril/valsartan vs valsartan on CV death and total HF hospitalizations), there was even some suggestion of greater benefit for sacubitril/valsartan among those with lower baseline kidney function.¹⁶

In the present analyses, although we observed some attenuation of the relative effect estimates for finerenone vs placebo among patients in lower categories of baseline eGFR, there was no definitive statistical heterogeneity in either the categorical or continuous analyses of baseline eGFR, whereas the absolute rate differences between treatment arms were consistent across the baseline eGFR categories. Similar patterns of treatment effect were noted in post hoc analyses of the Americas population of the TOPCAT trial^{17,18} that tested the steroidal MRA spironolactone (vs placebo) for the composite of death from CV causes, aborted cardiac arrest, or hospitalization for HF.⁶ In our present sensitivity analyses, relative treatment effects using competing risk and “model-free” approaches were also broadly similar

TABLE 5 Treatment Discontinuation by Randomized Arm According to Baseline CKD and UACR Categories

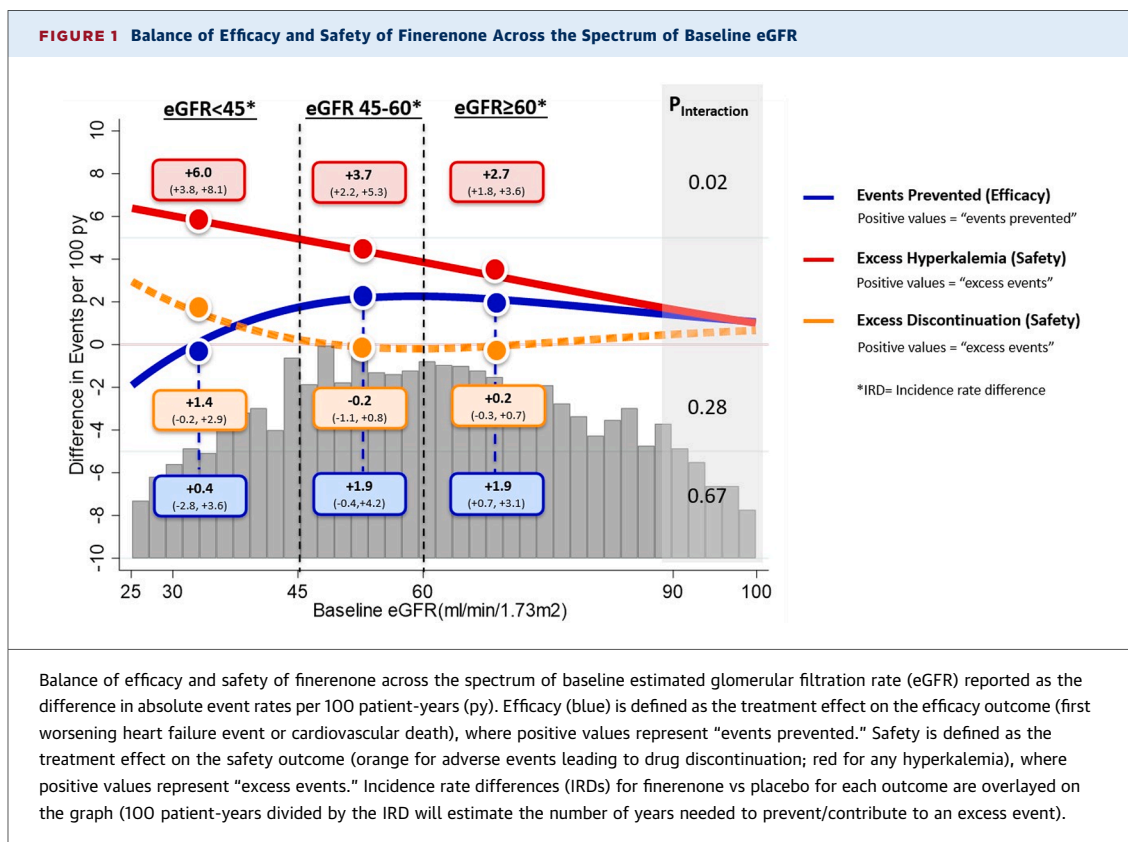
	Randomization Arm						<i>P</i> _{interaction}
	Placebo	Finerenone	Placebo	Finerenone	Placebo	Finerenone	
eGFR, mL/min/1.73 m ²	≥60		45 to <60		<45		
Any discontinuation							
Events/patients (%)	286/1,556 (18.4)	255/1,547 (16.5)	157/752 (20.9)	168/800 (21.0)	173/682 (25.4)	188/646 (29.1)	
Rate/100 py (95% CI)	7.9 (7.0-8.9)	7.0 (6.2-7.9)	9.5 (8.0-11.0)	9.5 (8.0-10.9)	12.6 (10.6-14.5)	15.2 (12.9, 17.4)	
HR (95% CI)	0.88 (0.74-1.04)		0.97 (0.78-1.21)		1.22 (0.99-1.50)		0.05
Absolute rate difference	−0.9 (−2.2 to 0.4)		−0.0 (−2.1 to 2.0)		2.6 (−0.3 to 5.6)		0.10
AEs leading to discontinuation							
Events/patients (%)	45/1,556 (2.9)	52/1,547 (3.4)	35/752 (4.7)	35/800 (4.4)	48/682 (7.0)	60/646 (9.3)	
Rate/100 py (95% CI)	1.2 (0.9-1.6)	1.4 (1.0-1.8)	2.1 (1.4-2.8)	2.0 (1.3-2.6)	3.5 (2.5-4.5)	4.8 (3.6-6.1)	
HR (95% CI)	1.13 (0.76-1.69)		0.90 (0.56-1.44)		1.40 (0.95-2.05)		0.35
Absolute rate difference	0.2 (−0.3 to 0.7)		−0.1 (−1.1 to 0.8)		1.4 (−0.2 to 2.9)		0.28
SAEs leading to discontinuation							
Events/patients (%)	23/1,556 (1.5)	23/1,547 (1.5)	11/752 (1.5)	15/800 (1.9)	18/682 (2.6)	14/646 (2.2)	
Rate/100 py (95% CI)	0.6 (0.4-0.9)	0.6 (0.4-0.9)	0.7 (0.3-1.1)	0.8 (0.4-1.3)	1.3 (0.7-1.9)	1.1 (0.5-1.7)	
HR (95% CI)	0.99 (0.55-1.76)		1.19 (0.54-2.61)		0.88 (0.43-1.78)		0.81
Absolute rate difference	−0.0 (−0.4 to 0.4)		0.2 (−0.4 to 0.8)		−0.2 (−1.0 to 0.7)		0.78
UACR, mg/g	<30		>30 to <300		≥300		
Any discontinuation							
Events/patients (%)	317/1,741 (18.2%)	308/1,757 (17.5)	198/867 (22.8)	207/843 (24.6)	76/281 (27.0)	76/292 (26.0)	
Rate/100 py (95% CI)	7.9 (7.0-8.8)	7.5 (6.7-8.4)	10.5 (9.0-12.0)	11.7 (10.1-13.4)	13.7 (10.6-16.8)	13.4 (10.2-16.5)	
HR (95% CI)	0.95 (0.81-1.11)		1.14 (0.94-1.39)		0.91 (0.66-1.25)		0.32
Absolute risk difference	−0.4 (−1.6 to 0.8)		1.2 (−1.0 to 3.5)		−0.4 (−4.7 to 4.0)		0.44
AEs leading to discontinuation							
Events/patients (%)	62/1,741 (3.6)	73/1,757 (4.2)	39/867 (4.5)	41/843 (4.9)	21/281 (7.5)	27/292 (9.2)	
Rate/100 py (95% CI)	1.6 (1.2-1.9)	1.8 (1.4-2.2)	2.1 (1.4-2.7)	2.3 (1.6-3.0)	3.8 (2.2-5.4)	4.7 (2.9-6.6)	
HR (95% CI)	1.16 (0.82-1.63)		1.17 (0.75-1.81)		1.24 (0.69-2.22)		0.99
Absolute risk difference	0.2 (−0.3 to 0.8)		0.3 (−0.7 to 1.2)		1.0 (−1.5 to 3.4)		0.85
SAEs leading to discontinuation							
Events/patients (%)	27/1,741 (1.6)	23/1,757 (1.3)	15/867 (1.7)	13/843 (1.5)	8/281 (2.8)	13/292 (4.5)	
Rate/100 py (95% CI)	0.7 (0.4-0.9)	0.6 (0.3-0.8)	0.8 (0.4-1.2)	0.7 (0.3-1.1)	1.4 (0.4-2.5)	2.3 (1.0-3.5)	
HR (95% CI)	0.84 (0.48-1.46)		0.97 (0.46-2.04)		1.64 (0.64-4.17)		0.51
Absolute risk difference	−0.1 (−0.5 to 0.2)		−0.1 (−0.6 to 0.5)		0.8 (−0.8 to 2.4)		0.52

AE = adverse event; SAE = serious adverse event; other abbreviations as in Tables 1, 2, and 4.

across baseline eGFR categories, although we also observed consistency in treatment effects across categories of baseline UACR, another dimension of kidney function.

Potential concerns of patients and clinicians regarding the use of MRAs include worsening indices of kidney function and the risk of hyperkalemia. We previously reported that both these AEs were more frequent among patients with lower baseline eGFR and were more common among those randomized to finerenone compared with placebo.^{19,20} Similar observations were also noted in the post hoc analyses of TOPCAT Americas,⁶ suggesting that patients with lower eGFR treated with MRAs should be carefully and regularly monitored. Conversely, the risk of

hypokalemia is lowered with finerenone use compared with placebo. This observation has particular relevance for patients with HFmrEF/HFpEF, where hypokalemia has been independently associated with adverse outcomes.²¹ In our present analyses we noted that the frequency of randomized treatment discontinuation was numerically higher for finerenone vs placebo among patients in the lowest category of baseline eGFR. The absolute risks of hyperkalemia also appeared to be amplified at lower eGFR. These observations suggest the need for careful monitoring to optimize individual risk-benefit ratios, especially among patients with more advanced kidney disease. Reassuringly, discontinuation of the randomized treatment due to AEs or



SAEs was relatively infrequent and similar across both baseline categories of eGFR and UACR. We acknowledge that FINEARTS-HF excluded patients with more advanced CKD (eGFR <25 mL/min/1.73 m²) and higher baseline serum potassium (>5 mmol/L), which are both risk factors for adverse outcomes, thus limiting extrapolation of the observed findings to such patients.

Prior analyses examining KDIGO (Kidney Disease: Improving Global Outcomes) risk categories (which combine eGFR and UACR data) did not find evidence of differential treatment effects on the main CV outcomes according to the baseline KDIGO risk.²² The present prespecified analyses explore treatment effects according to baseline eGFR and UACR separately. As such, we provide additional clinical insights according to these distinct biomarkers of kidney and CV risk. Although no clear evidence for heterogeneity of treatment effects across eGFR was observed, we recognize that there may be power limitations with respect to the relatively modest proportion of patients with more advanced CKD. Although beyond the scope of the current analyses, the fact that (per protocol) finerenone dosing was

lower among those with lower eGFR could also have relevance to the overall risk-benefit ratio. On the one hand, perhaps higher doses among those with lower eGFR could have resulted in a further reduction in CV events; however, on the other hand, whether this would have led to excess safety events (particularly hyperkalemia) would have to be carefully considered. Future analyses of the influence of dose on outcomes, including detailed pharmacokinetic analyses, will help address these questions.

STUDY STRENGTHS AND LIMITATIONS. Strengths of our analyses include the availability of robustly collected data and adjudicated outcomes across a broad spectrum of baseline kidney function in the setting of a randomized controlled trial. However, there are several limitations worth noting. Patients with more advanced kidney disease (eGFR <25 mL/min/1.73 m²) and serum potassium >5.0 mmol/L were excluded from the trial, whereas the overall proportion of patients with baseline eGFR ≥25 and <45 mL/min/1.73 m² was modest (22%). No adjustment for multiple testing was considered in the present analyses, raising the

possibility of false positive results. Finally, these findings may not be fully generalizable to patient populations with characteristics distinct from those enrolled in the FINEARTS-HF trial.

CONCLUSIONS

The benefits of finerenone to reduce the composite outcome of CV death and total HF events among patients with HFmrEF/HFpEF in FINEARTS-HF were generally consistent across baseline eGFR and UACR. Although absolute risks of hyperkalemia were amplified at lower eGFR, the risk of discontinuation due to AEs was consistent across eGFR categories. Given the overall favorable risk-to-benefit profile, baseline kidney function should not detract from the initiation of finerenone, but patients with lower eGFR will likely require more frequent monitoring and appropriate dose adjustment to obtain optimal CV benefits.

DATA SHARING Data are made available to qualified scientific and medical researchers through vivli.org. All requests will be reviewed by an independent scientific review panel and data provided according to the conditions laid out on <https://vivli.org/ourmember/bayer/>.

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ADDRESS FOR CORRESPONDENCE: Dr Finnian R. Mc Causland, Renal Division, Brigham and Women's Hospital, 75 Francis Street, MRB-4, Boston, Massachusetts 02115, USA. E-mail: fmccausland@bwh.harvard.edu.

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE:

Patients with HF often have concomitant CKD, which is associated with worse clinical outcomes and may lead to hesitancy to use therapies that alter kidney function. In the present study, we observed that among patients with HFmrEF/HFpEF, the effects of finerenone to reduce the composite outcome of CV death and total HF events were generally consistent across baseline eGFR and UACR.

TRANSLATIONAL OUTLOOK: Further studies exploring the implications of finerenone dose on efficacy and safety at lower levels of kidney function are required to inform optimal management of patients with coexisting HF and CKD.

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KEY WORDS chronic kidney disease, heart failure, kidney function

APPENDIX For supplemental figures and tables, please see the online version of this paper.